SIRURI DE NUMERE REALE. PROBLEME PROPUSE

1. Stabiliţi mărginirea şi monotonia şirului $a_n = \frac{1\cdot 3\cdot 5\cdot \ldots \cdot (2n-1)}{2\cdot 4\cdot 6\cdot \ldots (2n)}$ şi apoi calculaţi $\lim\limits_{n\to\infty} a_n$.

2. Să se demonstreze că şirul $a_n = \frac{n^n}{(n!)^2}, n \in \mathbb{N}^*$ este convergent şi să se calculeze $\lim\limits_{n\to\infty} a_n$.

   Calculaţi limitele:

3. $\lim\limits_{n\to\infty} \dfrac{n^k}{a^n}, \ \ k \in \mathbb{N}^*, \ \ a > 0$

4. $\lim\limits_{n\to\infty} \dfrac{\alpha^n}{n!}, \ \ \alpha > 0$

5. $\lim\limits_{n\to\infty} \sqrt[3]{n} \cdot (\sqrt[3]{(n+1)^2} - \sqrt[3]{(n-1)^2})$

6. $\lim\limits_{n\to\infty} n^k \cdot \left(\sqrt{\dfrac{n+2}{n+5}} - 1\right), \ \ k \in \mathbb{R}$

7. $\lim\limits_{n\to\infty} \sum\limits_{k=1}^{n} \dfrac{\sin k}{n^2+k}$

8. $\lim\limits_{n\to\infty} \dfrac{1 + 2\sqrt{2} + 3\sqrt{3} + \ldots + n\sqrt{n}}{n^2\sqrt{n}}$

9. $\lim\limits_{n\to\infty} \sin(\sqrt[3]{n^3+n+x} \cdot \pi)$

10. $\lim\limits_{n\to\infty} \sin(n\pi \sqrt[n]{\mathrm{e}})$

11. $\lim\limits_{n\to\infty} \sqrt{n} \left[\mathrm{e} - \left(1 + \dfrac{1}{n}\right)^n\right]$

12. $\lim\limits_{n\to\infty} \left(\dfrac{1}{2n+1} + \dfrac{1}{2n+2} + \ldots + \dfrac{1}{3n}\right)$

13. $\lim\limits_{n\to\infty} \sqrt[n]{\dfrac{(n!)^2}{(2n)!}}$

14. $\lim\limits_{n\to\infty} \sqrt[n]{\dfrac{3^{3n}(n!)^3}{(3n)!}}$

15. $\lim\limits_{n\to\infty} \sum\limits_{k=1}^{n} \dfrac{k^2+k-1}{(k+1)!}$

16. $\lim\limits_{n\to\infty}\sum\limits_{k=1}^{n}\dfrac{2^{k-1}(k-1)}{(k+1)!}$

17. $\lim\limits_{n\to\infty}\sum\limits_{k=1}^{n}\left(\sqrt{1+\dfrac{k}{n^2}}-1\right)$

18. $\lim\limits_{n\to\infty}\left(1-\dfrac{1}{3}\right)\left(1-\dfrac{1}{6}\right)\left(1-\dfrac{1}{10}\right)\ldots\left(1-\dfrac{1}{C_{n+1}^2}\right)$

19. $\lim\limits_{n\to\infty}\dfrac{C_n^1\cdot C_n^2\cdot C_n^3\cdot\ldots\cdot C_n^p}{n^{\frac{p(p+1)}{2}}},\ \ p\in\mathbb{N}^*, p\leq n$

20. $\lim\limits_{n\to\infty}\dfrac{2^2\cdot 4^2\cdot 6^2\cdot\ldots\cdot(2n)^2}{1^2\cdot 3^2\cdot 5^2\cdot\ldots\cdot(2n-1)^2}\cdot\dfrac{1}{\sqrt[n]{n!}}$